

Multilingual Question-Answer (QnA) Generation System

Project Output Evidence & Demonstration Screenshots

This document presents the supplied screenshots of the Streamlit interface, processing pipeline, generated multilingual Q&A; results, configuration options, and final Excel worksheets.

Output formats demonstrated: PDF input → English / Hindi / Marathi Q&A; → QnA.xlsx

1. Application Interface & Input Processing

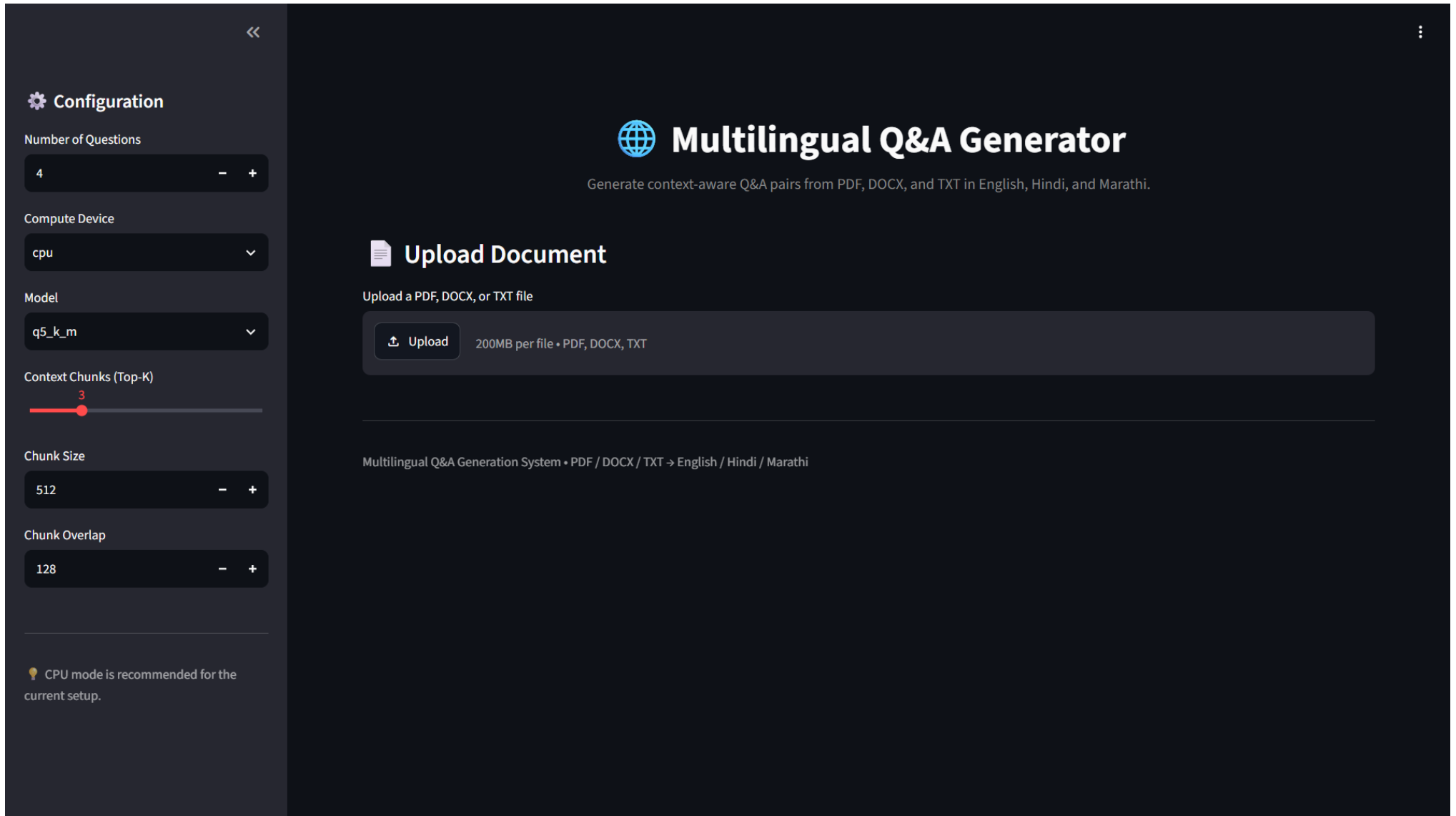


Figure 1 — Initial Streamlit Interface

Configuration

Number of Questions

4

Compute Device

cpu

Model

q5_k_m

Context Chunks (Top-K)

3

Chunk Size

512

Chunk Overlap

128

CPU mode is recommended for the current setup.

Upload Document

Upload a PDF, DOCX, or TXT file

IKS.pdf170.4KB

Selected file: IKS.pdf

File type: .PDF

Generate Q&A

Pipeline Progress

Generating Q&A... 18s

2026-09-26 22:40:35,351 - src.embeddings - INFO - Loading embedding model: sentence-transformers/all-MiniLM-L6-v2

2026-09-26 22:40:35,351 - sentence_transformers.SentenceTransformer - INFO - Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2

2026-09-26 22:40:36,251 - sentence_transformers.SentenceTransformer - INFO - Use pytorch device: cpu

2026-09-26 22:40:36,253 - src.embeddings - INFO - Embedding model loaded on cpu

2026-09-26 22:40:37,928 - src.embeddings - INFO - Generated embeddings for 65 texts

2026-09-26 22:40:37,934 - faiss.loader - INFO - Loading faiss with AVX2 support.

2026-09-26 22:40:37,934 - faiss.loader - INFO - Could not load library with AVX2 support due to: ModuleNotFoundError("No module named 'faiss.swigfaiss_avx2'")

2026-09-26 22:40:37,934 - faiss.loader - INFO - Loading faiss.

2026-09-26 22:40:38,234 - faiss.loader - INFO - Successfully loaded faiss.

2026-09-26 22:40:38,246 - src.embeddings - INFO - Built FAISS index with 65 chunks

2026-09-26 22:40:38,246 - src.pipeline - INFO - \u2713 Generated embeddings and built vector store

2026-09-26 22:40:38,246 - src.pipeline - INFO - [TIMING] Step 5 - Embeddings + FAISS: 16.29s

2026-09-26 22:40:38,246 - src.pipeline - INFO - Step 6: Generating 4 Q&A pairs

2026-09-26 22:40:38,246 - src.embeddings - INFO - Loading embedding model: sentence-transformers/all-MiniLM-L6-v2

2026-09-26 22:40:38,247 - sentence_transformers.SentenceTransformer - INFO - Load pretrained SentenceTransformer: sentence-transformers/all-MiniLM-L6-v2

2026-09-26 22:40:38,350 - sentence_transformers.SentenceTransformer - INFO - Use pytorch device: cpu

2026-09-26 22:40:38,351 - src.embeddings - INFO - Embedding model loaded on cpu

2026-09-26 22:40:38,368 - src.embeddings - INFO - Generated embeddings for 1 texts

2026-09-26 22:40:38,373 - src.qna_generator - INFO - Generating Q&A (attempt 1/2)

Figure 2 — PDF Upload and Q&A; Generation in Progress

Configuration

Number of Questions

4

-

+

Compute Device

cpu

Model

q5_k_m

Context Chunks (Top-K)

3

Chunk Size

512

-

+

Chunk Overlap

128

-

+

CPU mode is recommended for the current setup.

2026-09-26 22:43:31,090 - __main__ - INFO - Pipeline completed successfully.

2026-09-26 22:43:31,090 - __main__ - INFO - \u2713 Q&A generation complete!

2026-09-26 22:43:31,090 - __main__ - INFO - \u2713 Output saved to: output\QnA.xlsx

Q&A generated successfully in 192 seconds!

Generated Q&A

GB English

IN Hindi

IN Marathi

Questions	Answers
What disciplines does Indian Knowledge Systems (IKS) encompass?	science, philosophy, medicine, mathematics, arts, and environmental management
What global challenges does reviving Indian Knowledge Systems (IKS) aim to address?	health, environment, and social harmony
What subjects do the Shastras cover, and how are they characterized?	The Shastras cover subjects from logic, ethics, and linguistics to architecture, astronomy, a
What is the core of Indian Knowledge Systems according to the text?	The core of Indian Knowledge Systems lies in the Vedas and Vedangas.

Total Q&A pairs: 4

Download Result

Download QnA.xlsx

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Pipeline Logs

Multilingual Q&A Generation System • PDF / DOCX / TXT → English / Hindi / Marathi

Figure 3 — Generated English Q&A; Output

 Configuration

Number of Questions

4

Compute Device

cpu

Model

q5_k_m

Context Chunks (Top-K)

3

Chunk Size

512

Chunk Overlap

128

 CPU mode is recommended for the current setup.

2026-09-26 22:43:31,090 - __main__ - INFO - Pipeline completed successfully.

2026-09-26 22:43:31,090 - __main__ - INFO - \u2713 Q&A generation complete!

2026-09-26 22:43:31,090 - __main__ - INFO - \u2713 Output saved to: output\QnA.xlsx

🎉 Q&A generated successfully in 192 seconds!

 Generated Q&A

GB EnglishIN HindiIN Marathi

Questions	Answers
भारतीय ज्ञान प्रणाली (आईकेएस) कोणते अनुशासनांना समावेश आहे?	विज्ञान, दर्शन, औषध, गणित, कला आणि पर्यावरण प्रबंधन
कोणते वैश्विक चुनौतियां भारतीय ज्ञान प्रणाली (IKS) लक्ष देण्यासाठी पुनर्जीवित करतात?	स्वास्थ्य, पर्यावरण आणि सामाजिक संतुलन
शॅस्ट्रांना कोणते विषय कवरतात, आणि ते कसे वर्णन करतात?	शस्त्रास तार्किक, नैतिकता आणि भाषाविज्ञानपासून आर्किटेक्चर, खगोलशास्त्र आणि शासनपर्यंत विषयांना क
भारतीय ज्ञान प्रणालीचे आधार काय आहे?	भारतीय ज्ञान प्रणालीचे कोर Vedas आणि Vedangasमध्ये आहे.

Total Q&A pairs: 4

↓ Download Result

 Download QnA.xlsx

> 📄 Pipeline Logs

Multilingual Q&A Generation System • PDF / DOCX / TXT → English / Hindi / Marathi

Figure 4 — Generated Marathi Q&A; Output

Configuration

Number of Questions

4

Compute Device

cpu

Model

q5_k_m

Context Chunks (Top-K)

3

Chunk Size

512

Chunk Overlap

128

💡 CPU mode is recommended for the current setup.

Download Result

Download QnA.xlsx

Pipeline Logs

```
2026-09-26 22:40:20,219 - __main__ - INFO - Starting Q&A generation pipeline...
2026-09-26 22:40:20,219 - src.pipeline - INFO - =====
2026-09-26 22:40:20,219 - src.pipeline - INFO - Starting Q&A Generation Pipeline
2026-09-26 22:40:20,219 - src.pipeline - INFO - =====
2026-09-26 22:40:20,219 - src.pipeline - INFO - Step 1: Extracting text from input\IKS.pdf
2026-09-26 22:40:20,767 - src.document_processor - INFO - Extracted 28656 characters from PDF
2026-09-26 22:40:20,769 - src.pipeline - INFO - \u2713 Extracted 28650 characters
2026-09-26 22:40:20,769 - src.pipeline - INFO - [TIMING] Step 1 - Document extraction: 0.55s
2026-09-26 22:40:20,769 - src.pipeline - INFO - Step 2: Detecting language
2026-09-26 22:40:20,771 - src.pipeline - INFO - \u2713 Detected language: english
2026-09-26 22:40:20,771 - src.pipeline - INFO - [TIMING] Step 2 - Language detection: 0.00s
2026-09-26 22:40:20,771 - src.pipeline - INFO - Step 3: Initializing models
2026-09-26 22:40:20,772 - src.model_manager - INFO - Model found at models\Qwen3-4B-Q5_K_M.gguf
2026-09-26 22:40:20,773 - src.qna_generator - INFO - Loading LLM from: models\Qwen3-4B-Q5_K_M.gguf
2026-09-26 22:40:21,952 - src.qna_generator - INFO - LLM loaded successfully on cpu
2026-09-26 22:40:21,952 - src.translator - INFO - Using M2M100 translator for Hindi/Marathi translation
2026-09-26 22:40:21,952 - src.pipeline - INFO - \u2713 Models initialized
2026-09-26 22:40:21,952 - src.pipeline - INFO - [TIMING] Step 3 - Model initialization: 1.18s
2026-09-26 22:40:21,952 - src.pipeline - INFO - Step 4: Chunking text (size=512, overlap=128)
2026-09-26 22:40:21,953 - src.chunker - INFO - Created 65 chunks (sentence-based)
2026-09-26 22:40:21,953 - src.pipeline - INFO - \u2713 Created 65 chunks
2026-09-26 22:40:21,953 - src.pipeline - INFO - [TIMING] Step 4 - Text chunking: 0.00s
2026-09-26 22:40:21,953 - src.pipeline - INFO - Step 5: Generating embeddings
2026-09-26 22:40:35,351 - src.embeddings - INFO - Loading embedding model: sentence-transformers/all-MiniLM-L6-v2
2026-09-26 22:40:35,351 - sentence_transformers.SentenceTransformer - INFO - Load pretrained SentenceTransformer: sentenc
2026-09-26 22:40:36,251 - sentence_transformers.SentenceTransformer - INFO - Use pytorch device: cpu
2026-09-26 22:40:36,253 - src.embeddings - INFO - Embedding model loaded on cpu
```

Figure 5 — Pipeline Logs and Processing Steps

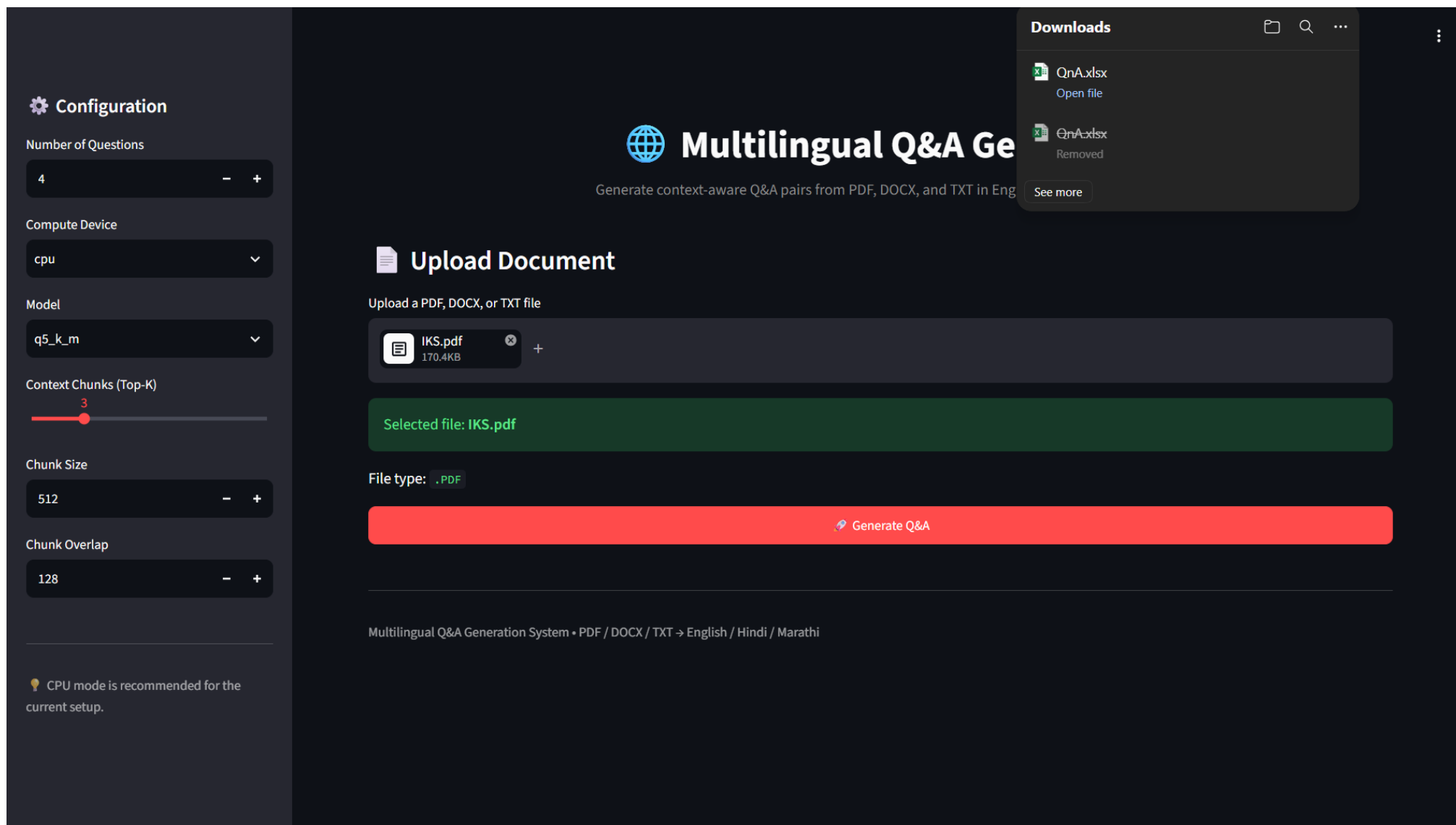


Figure 6 — Selected Input and Downloaded Excel Result

Configuration

Number of Questions

4

Compute Device

cpu

Model

q5_k_m

Context Chunks (Top-K)

3

Chunk Size

512

Chunk Overlap

128

CPU mode is recommended for the current setup.

Multilingual Q&A Generator

Generate context-aware Q&A pairs from PDF, DOCX, and TXT in English, Hindi, and Marathi.

Upload Document

Upload a PDF, DOCX, or TXT file

IKS.docx19.1KB

Selected file: IKS.docx

File type: .DOCX

Generate Q&A

Pipeline Progress

Generating Q&A... 7s

2026-09-26 22:46:30,813 - src.embeddings - INFO - Loading embedding model: sentence-transformers/all-MiniLM-L6-v2
2026-09-26 22:46:30,813 - sentence_transformers.SentenceTransformer - INFO - Load pretrained SentenceTransformer: sentence-tr
2026-09-26 22:46:31,261 - sentence_transformers.SentenceTransformer - INFO - Use pytorch device: cpu
2026-09-26 22:46:31,262 - src.embeddings - INFO - Embedding model loaded on cpu
2026-09-26 22:46:32,361 - src.embeddings - INFO - Generated embeddings for 52 texts
2026-09-26 22:46:32,370 - faiss.loader - INFO - Loading faiss with AVX2 support.
2026-09-26 22:46:32,370 - faiss.loader - INFO - Could not load library with AVX2 support due to:

Figure 7 — Q&A; Generation Interface and Processing View

2. Configuration & UI Options

Compute Device

cpu



cpu

cuda

auto

Figure 8 — Compute Device Options

Model

q5_k_m



q5_k_m

q4_k_m

Figure 9 — Model Selection Options

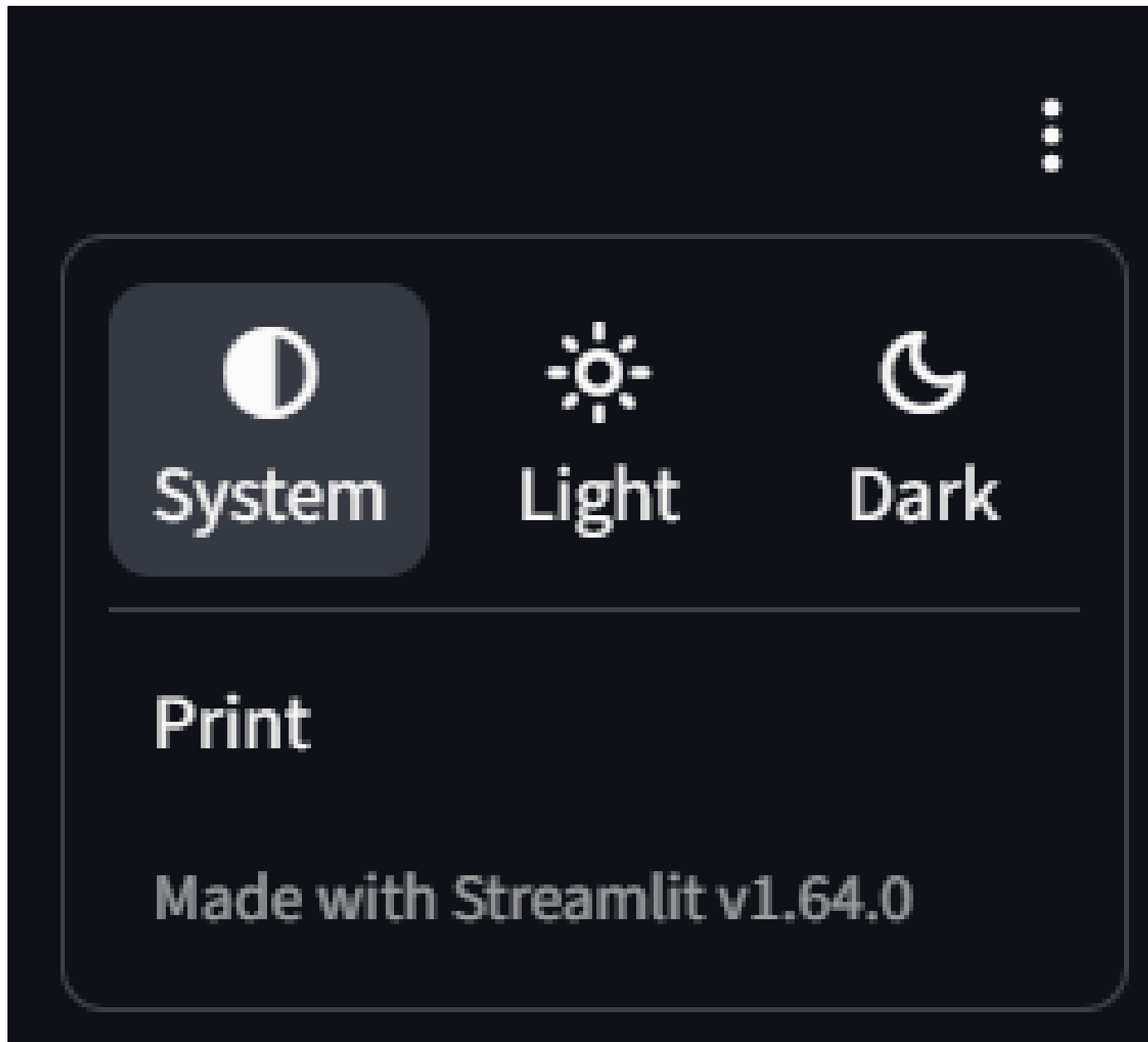


Figure 10 — Streamlit Display and Theme Menu

3. Final Excel Output

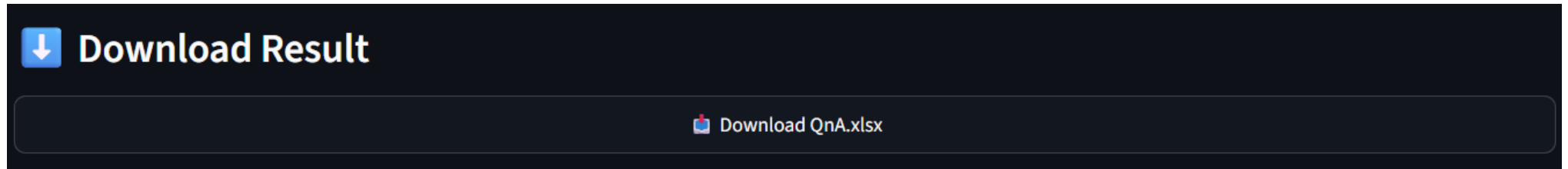


Figure 11 — QnA.xlsx Download Result

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Figure 12 — English Worksheet Output

A1	Questions
Questions	Answers
1 भारतीय ज्ञान प्रणालियों (आईकेएस) में कौन सी अनुशासन शामिल हैं?	विज्ञान, दर्शन, चिकित्सा, गणित, कला, और पर्यावरण प्रबंधन
2 भारतीय ज्ञान प्रणालियों (आईकेएस) को पुनर्जीवित करने के लिए क्या वैश्विक चुनौतियों का सामना करना पड़ता है?	स्वास्थ्य, पर्यावरण और सामाजिक सामंजस्य
3 Shastras किस विषयों को कवर करते हैं, और वे कैसे वर्णित होते हैं?	Shastras तार्किक, नैतिकता और भाषाविज्ञान से लेकर वास्तुकला, खगोल विज्ञान और शासन तक विषयों को कवर करता है। ये प्रणालियां दोनों सैद्धांतिक और अत्यधिक व्यावहारिक हैं, जैसा कि दैनिक जीवन, स्वास्थ्य देखभाल, इंजीनियरिंग और शिक्षा में उनके अनुप्रयोगों में देखा गया है।
4 पाठ के अनुसार भारतीय ज्ञान प्रणाली का कोर क्या है?	भारतीय ज्ञान प्रणालियों का कोर वेडा और वेडांग्स में स्थित है।
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Figure 13 — Hindi Worksheet Output

A1	Questions
A	B
Questions	Answers
1 भारतीय ज्ञान प्रणाली (आईकेएस) कोणते अनुशासनांना समावेश आहे?	विज्ञान, दर्शन, औषध, गणित, कला आणि पर्यावरण प्रबंधन
2 कोणते वैश्विक चुनौतियां भारतीय ज्ञान प्रणाली (IKS) लक्ष देण्यासाठी पुनर्जीवित करतात?	स्वास्थ्य, पर्यावरण आणि सामाजिक संतुलन
3 शॅस्ट्रांना कोणते विषय कवरतात, आणि ते कसे वर्णन करतात?	शस्त्रास ताकिक, नैतिकता आणि भाषाविज्ञानपासून आर्किटेक्चर, खगोलशास्त्र आणि शासनपर्यंत विषयांना कवरतात. हे सिस्टम सिद्धांतिक आणि अत्यंत व्यावहारिक आहेत, जे त्यांच्या दैनिक आयुष्यात, आरोग्य, इंजीनियरिंग, आणि शिक्षण अनुप्रयोगमध्ये दिसते.
4 भारतीय ज्ञान प्रणालीचे आधार काय आहे?	भारतीय ज्ञान प्रणालीचे कोर Vedas आणि Vedangasमध्ये आहे.
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Figure 14 — Marathi Worksheet Output